Reproducible Experiments

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Responsible Plant Classifier

Before you start

AI-2 Week 35: comparisons need identifiable versions and fixed criteria.

Plan time to read, investigate and explain. Keep predictions separate from observations.

Your mission

> Start with the target: Record enough detail to reproduce a run. Try the recall before opening notes.

Your learning target

I can record enough detail to reproduce a run.

I can distinguish **reproducibility** from validity.

I can report all planned **seed**s instead of selecting the best.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Metrics Match Consequences. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Hold Out Test Data. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Mean, Median, and Baselines. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

A result is reproducible only when another builder can recreate the data, split, code, settings, **seed**, and metric calculation.



Your target

LEARNING TARGET Two teams use the same **random seed** but different label files. Their results differ. List the records needed to reproduce the experiment.

Builder name and experiment date:

Readiness for the connected explanation

What must accompany a reported validation score?

Readiness for Versioned experiments

Why should expected test results be written before running the function?

My readiness explanation

This week’s end goal

By the end, I can explain reproducible experiments using a traced example and a stated limitation.

Notice the evidence

> Investigate the supplied Reproducible Experiments case. Separate observations, given facts and assumptions before choosing an explanation.

PICTURE ENTRY

Notice → Predict → Explain

Begin with visible **evidence**; do not explain more than the picture and information support.



Start with this puzzle

MYSTERY The team must use reproducible experiments to solve a model-building problem. What information do they have? What important **evidence** is still missing?

Name three visible details without interpreting them.

Predict the data, model, or evaluation decision being made.

Identify one hidden assumption that could change the result.

State what **evidence** would confirm or reject your **prediction**.

My **prediction** and supporting **evidence**:

A closer explanation

A **seed** can help repeat a random split or initialization, but it is not a magic quality stamp. Save **data version**, split membership, code **version**, settings, pre**process**ing, **seed** and metric definition.

Some software operations remain nondeterministic, so document tolerances and environment too.

Use fictional classroom data throughout this investigation. Keep the task, units, reference labels and data split visible as you reason.

Earlier weeks supplied the model-building **process**; this week examines experimental records and **seed**s.

Apply the connected idea: Versioned experiments

A fictional activity scoring experiment compares two weight settings on the same development records. Save code **version**, **input version**, weights, tie rule and **output**s.

A label such as v2 is useful only when it refers to an identifiable saved state.

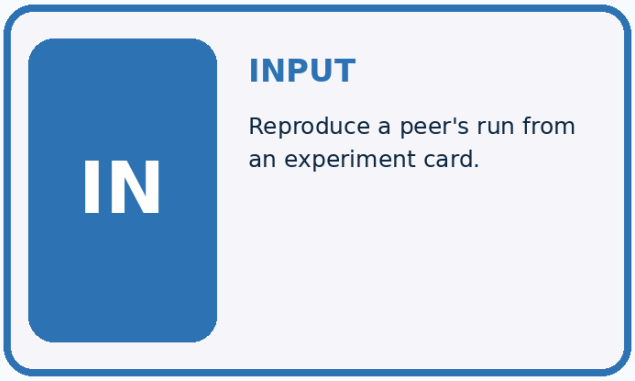
See how it works

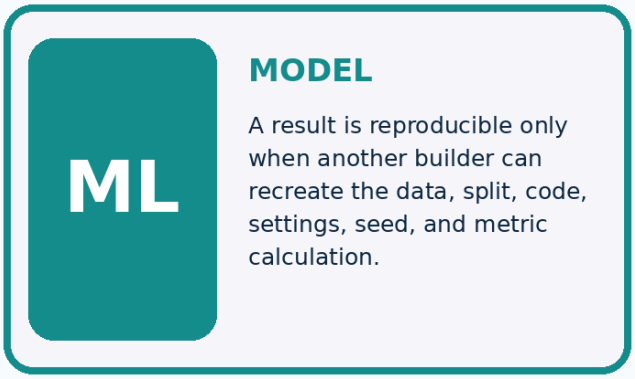
> Follow the mechanism for Reproducible Experiments. Connect each labelled input, deciding step and result.

ML MECHANISM

Input → Model → Evidence

Trace the information, fitted behavior, and evaluation proof separately.







Question to explain

QUESTION Reproduce a peer's run from an experiment card.

MODEL IDEA A result is reproducible only when another builder can recreate the data, split, code, settings, **seed**, and metric calculation.

PROOF NEEDED Use an unseen case, visible calculation, error analysis, and a stated limitation.

How the mechanism works

UNDER THE HOOD Record data=v2, split=grouped-s1, code=commit A7, **seed**=42. Record pre**process**ing, features, model type, settings, and library **version**s.

Draw a fresh data → model → **prediction** → evaluation trace:

A closer explanation

A team declares **seed**s 11, 22, 33 before running. Validation accuracies are 14/20, 16/20, 12/20:70, 80, 60 percent. Mean accuracy is 70 percent, with a 20-percentage-point range.

Reporting only **seed** 22 as the model result hides the declared experiment.

Before continuing, underline the decision rule and circle the **evidence** that supports the conclusion. Explain the example to a partner without replacing the counts or conditions with a stronger claim.

Keep the distinction between an observed result and a proposed next step.

Apply the connected idea: Versioned experiments

```
items = [{"name": "A", "fun": 4, "learning": 1},
         {"name": "B", "fun": 2, "learning": 4}]
for version, fun_weight, learning_weight in [("v1", 2, 1), ("v2", 1, 2)]:
    scores = []
    for item in items:
        scores.append(fun_weight * item["fun"]
                      + learning_weight * item["learning"])
    print(version, scores)
```

Follow the evidence

STEP OR CASE	EVIDENCE	MEANING
v1 weights 2 and 1	A 9; B 8	A ranks first
v2 weights 1 and 2	A 6; B 10	B ranks first

Read each row from left to right and connect the evidence to the stated rule.

The ranking changes because the preference weights changed. That is a policy comparison, not proof that v2 is more accurate.

Keep the same data and constraints when comparing weights, and record why a policy better fits the stated goal.

Compare one planned change at a time

For the worked records, v1 gives A a score of 2 times 4 plus 1, or 9, and B a score of 2 times 2 plus 4, or 8. v2 gives A 4 plus 2 times 1, or 6, and B 2 plus 2 times 4, or 10. The winner changes because the weights changed.

Keep the records fixed so this explanation can be checked. A different preference policy is not automatically a more accurate model.

Connect the picture and the explanation

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

STUDENT LESSON · 4 OF 18

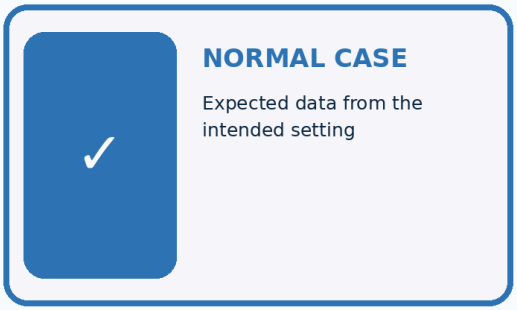
Compare the cases

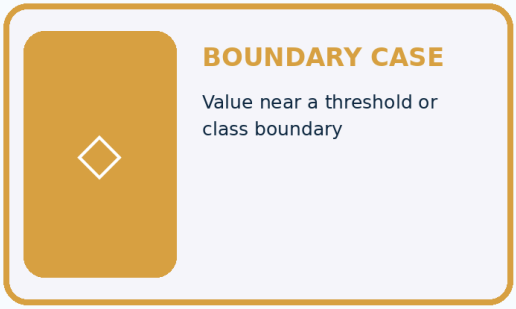
> Compare the supplied cases for Reproducible Experiments. Identify the detail that changes the conclusion.

EVALUATION DESIGN

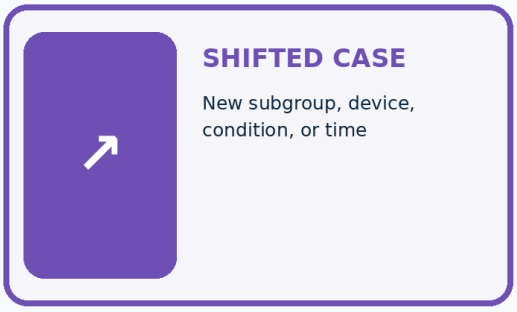
Four Tests, Four Kinds of Evidence

A system is not understood until it survives more than one comfortable example.









NORMAL Expected data and expected behavior.

Keep this boundary clear

BOUNDARY Value close to a threshold, ambiguous label, or uncertain **prediction**.

MISSING Necessary field, label, source, or measurement is unavailable.

SHIFTED New lighting, species, device, subgroup, or time condition.

RELEASE RULE Passing one case never proves generalization. Predefine expected behavior for all four categories.

Which case most threatens today's claim, and why?

Change one thing in the pictured case

Change one thing

Change: The report deletes the **60%** run.

Predicted result

Prediction: The remaining mean rises to75%, but the reported comparison becomes selectively incomplete.

Why the result follows

Reason: Removing an unfavourable declared run changes the **evidence** rather than improving the method.

STUDENT LESSON · 5 OF 18

Words that unlock the idea

> Use the vocabulary of Reproducible Experiments to explain a concrete decision, rather than only memorising definitions.

VOCABULARY

Words You Can Calculate and Defend

A definition matters when it lets you interpret a new dataset, model, or result.


1
REPRODUCIBILITY
ability to recreate a result
2
DATA VERSION
identified dataset state
3
RANDOM SEED
value controlling repeatable randomness
4
EXPERIMENT RECORD
complete inputs, settings, and outcomes

Term	Precise meaning	My ML example
reproducibility	ability to recreate a result	
data version	identified dataset state	
random seed	value controlling repeatable randomness	
experiment record	complete inputs, settings, and outcomes	

MISCONCEPTION TO REPAIR Writing down the final accuracy makes an experiment reproducible.

Repair the claim using a definition, calculation, or test record:

Words used in the closer explanation

Word	Meaning
Seed	An initial value controlling a pseudo-random process
Run record	A saved account of inputs, settings and outputs
Reproducibility	Ability to repeat a specified procedure
Cherry-picking	Reporting favorable results while hiding relevant others

Terms for Versioned experiments

Term	Meaning
Version	An identified state of code or rules
Configuration	Settings used for a run
Development set	Cases used while choosing or improving a method

Term	Meaning
Holdout	Cases reserved for an evaluation after choices are fixed

STUDENT LESSON · 6 OF 18

Follow a worked example

> Follow the worked trace for Reproducible Experiments, predict a step, then compare with its explanation.

WORKED EVIDENCE

Reason Step by Step

Predict first. Then check each step. A correct answer without visible **evidence** cannot be audited.

WEEK 26 • WORKED EVIDENCE

1. Record data=v2, split=grouped-s1, code=commit A7, seed=42.
2. Record preprocessing, features, model type, settings, and library versions.
3. Peer reruns and gets 16/20 correct, matching 80% accuracy.
4. A mismatch triggers environment and data comparison—not silent replacement.

Step 1

STEP 1 Record data=v2, split=grouped-s1, code=commit A7, **seed**=42.

Step 2

STEP 2 Record pre**process**ing, features, model type, settings, and library **version**s.

Step 3

STEP 3 Peer reruns and gets 16/20 correct, matching **80%** accuracy.

Step 4

STEP 4 A mismatch triggers environment and data comparison—not silent replacement.

CHECK Verify the denominator, unit, split, class definition, and **data version** before interpreting a score.

Recalculate, retrace, or restate the **evidence** independently:

A closer explanation

Experimental records and seeds at a glance

PART	EVIDENCE OR ACTION	INTERPRETATION
Before running	Declare 11 22 33	Save data and settings
Observed results	70 80 60 percent	Keep all three runs
Summary	Mean 70 percent	Range 20 points

The seed is part of the procedure, not evidence that a split represents the target population. Reproducing leakage perfectly reproduces a mistake. Check group separation and legitimate features before celebrating identical outputs.

A file contains only seed 22 and 80 percent. Name four missing items. Explain the steps or evidence behind your response.

Apply the connected idea: Versioned experiments

If the data also changes between v1 and v2, can the ranking change be attributed only to weights?

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Reproducible Experiments

Record variation across declared **seed**s instead of selecting the prettiest run.

The facts we will use

Supplied facts

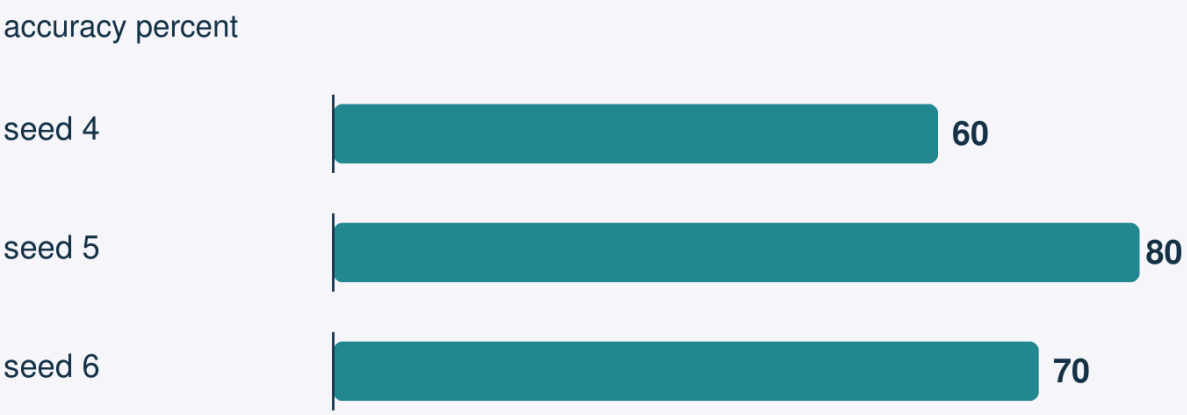
Same fixed validation set of10 cases; declared runs **seed** 4=6 correct, **seed** 5=8, **seed** 6=7. All three are reported.

Supplied facts

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

AI ACADEMY / AI-4/26

All declared seeds



Bar length shows the supplied value; the marked baseline is zero.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-4/26

Reproducible Experiments

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Keep run identities

Record seeds4, 5, 6 with 6/10, 8/10, 7/10.



2. Summarise variation

Mean accuracy is 70%; range is 60% to80%.



3. Preserve context

Record data version, split, code, settings and seeds, not just the winning80%.

CHANGE ONE THING

The report deletes the 60% run.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Keep run identities	Record seeds4, 5, 6 with 6/10, 8/10, 7/10.
2. Summarise variation	Mean accuracy is 70%; range is 60% to80%.
3. Preserve context	Record data version, split, code, settings and seeds, not just the winning80%.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

Read the picture in order. Record **data version**, split, code, settings and **seed**s, not just the winning**80%**. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

What this example does not establish

Keep the stated limit

Seeds help reproduction but do not necessarily remove hardware or library nondeterminism;

report the actual environment.

Keep the stated limit

Read the labelled picture: What should accompany the mean **70%**? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Keep the stated limit

Change one thing: The report deletes the **60%** run. Predict the result before checking the explanation. What remains the same?

STUDENT LESSON · 7 OF 18

Find and repair a mistake

> Find the unsupported step in the Reproducible Experiments claim and explain a repair using the supplied evidence.

Find the first error

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, data problem, or experiment error.

!

COMPARE • DIAGNOSE • REPAIR

Tempting claim: Writing down the final accuracy makes an experiment reproducible. Inspect the dataset, split, calculation, error type, subgroup, or experiment record; identify one testable cause; change one factor; rerun the same evidence.

Evidence record	Expected	Actual	Likely cause / next test
normal case	passes		
boundary case	defined behavior		
missing case	safe fallback		
shifted case	measured limit		

CLAIM TO CHALLENGE Writing down the final accuracy makes an experiment reproducible.

Diagnose one cause and name the cleanest next experiment:

A closer explanation

When comparing two methods, use the same declared splits or **seed**s where appropriate and compare paired results. A mean across reused samples does not automatically provide a statistical confidence claim.

Report the procedure and observed variation plainly.

> **One planned seed fails before prediction. How should it appear in the record?**

Apply the connected idea: Versioned experiments

Once **holdout** results guide another change, those cases have become development **evidence**. A later independent evaluation needs new reserved cases. Small **holdout**s still provide limited **evidence**.

> **You revise the policy after reading final-case failures. Are those cases still untouched holdout evidence?**

STUDENT LESSON · 8 OF 18

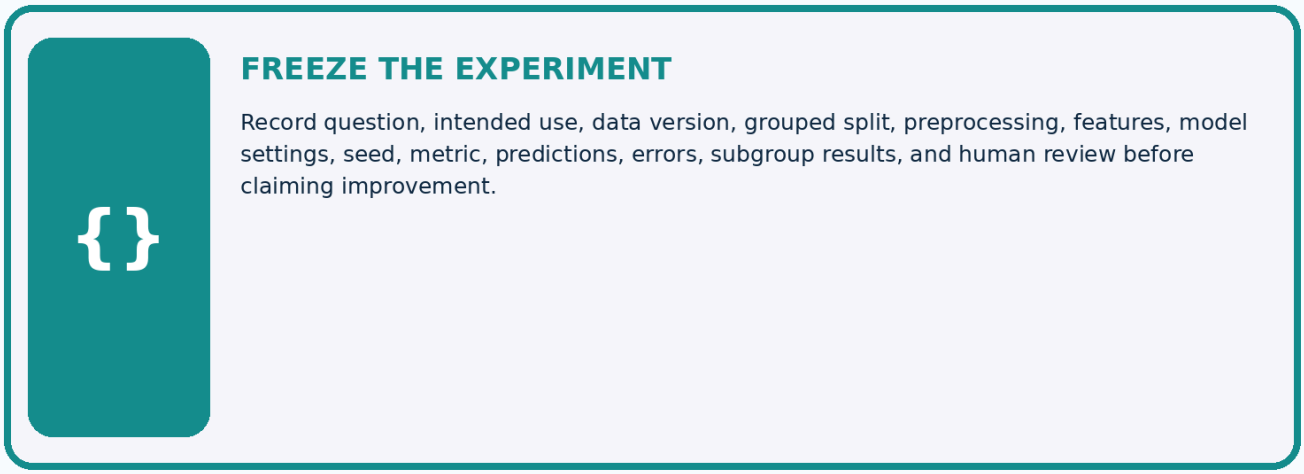
Practise together

> Practise Reproducible Experiments with support, then explain the deciding step in your own words.

ML STUDIO

Specify Before You Run

A trustworthy result can be reproduced from the recorded data, code, settings, and evaluation rule.



Experiment record

Question to explain

QUESTION + intended use + success threshold

DATA version + grouped split + leakage check

CODE/pre**process**ing + features + model + settings + **seed**

METRICS + positive class + units + denominators

PREDICTIONS + errors + subgroup slices + limitations

HUMAN reviewer + release, revise, limit, or stop decision

Write the minimum record needed to reproduce today's result:

A closer explanation

A small record can contain run ID, timestamp, dataset checksum, **training** and validation IDs, model settings, feature **version**, **seed**, **prediction**s and notes. Keep personal information out of exported logs.

Save failed runs with reasons instead of quietly removing them.

Apply the connected idea: Versioned experiments

Overwriting a file called latest removes the ability to reproduce earlier results unless **version**s are retained elsewhere. Save identifiable **version**s and a run log.

Avoid editing a reserved final test after seeing its **output** to make the chosen **version** look better.

Practise with less help: Improve a record

Use workbook W2 for the connected example “Experimental records and **seed**s”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for Experimental records and **seed**s. Proposed explanation: “A final answer is enough”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Practise the connected topic: Versioned experiments

Use its named workbook W2. Identify the given **input**s, try the middle steps with less help, then explain your trace. Check this claim before group discussion: A successful run proves the conclusion

Explain the picture

What should accompany the mean **70%**? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

My explanation using a pictured step and the given facts

This is supported practice. Keep the separately named independent case for an attempt before feedback.

STUDENT LESSON · 9 OF 18

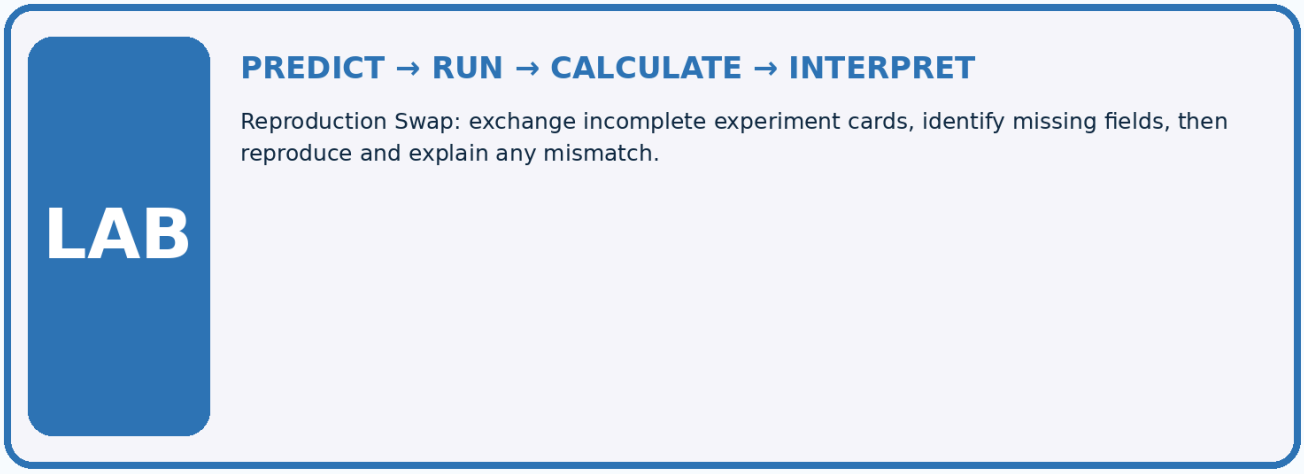
Run the investigation

> Investigate Reproducible Experiments. Record the prediction, actual observation or labelled paper trace, explanation and next test separately.

EVIDENCE LAB

Predict → Run → Calculate → Interpret

Keep the test conditions fixed, record every result, and change only one planned factor.



1. Write the hypothesis and expected **evidence** before running.

2. Reproduction Swap: exchange incomplete experiment cards, identify missing fields, then reproduce and explain any mismatch.

3. Run one normal, one boundary, one missing, and one shifted case.

4. Calculate the metric from raw counts or row-level errors.

5. Interpret what changed, what did not, and what remains unsafe or unknown.

Case · Prediction · Actual · Calculation / evidence · Meaning / next test

> **Case:** Normal
> **Prediction:** _____
> **Actual:** _____
> **Calculation / evidence:** _____
> **Meaning / next test:** _____

> **Case:** Boundary
> **Prediction:** _____
> **Actual:** _____
> **Calculation / evidence:** _____

> **Meaning / next test:** _____

> **Case:** Missing
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Calculation / evidence:** _____
>
> **Meaning / next test:** _____

> **Case:** Shifted
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Calculation / evidence:** _____
>
> **Meaning / next test:** _____

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

STUDENT LESSON · 10 OF 18

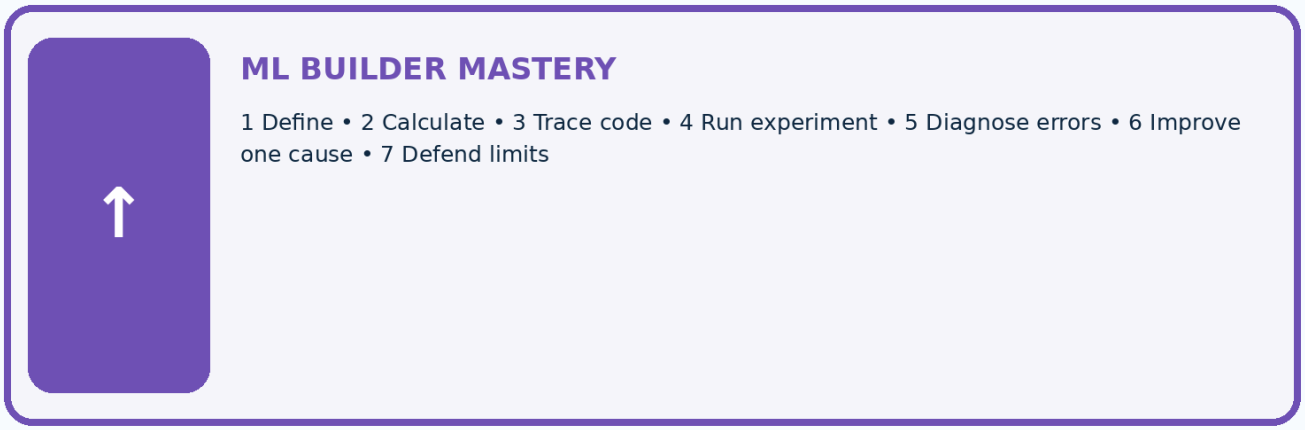
Build your understanding

> Move from recall to tracing, diagnosis and transfer in Reproducible Experiments. Explain why each step is justified.

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce **evidence** without copying the example.



DEFINE Use all four terms precisely: **reproducibility**, **data version**, **random seed**, **experiment record**.

CALCULATE Reproduce the worked **evidence** with units and denominator.

TRACE Explain the data, code or decision path.

DIAGNOSE Writing down the final accuracy makes an experiment reproducible.

TRANSFER Solve a fresh dataset or model case.

IMPROVE Change one cause and rerun fixed **evidence**.

DEFEND State intended use, metric, limitation, and human responsibility.

MASTERY RULE A coached calculation becomes mastery only after a different case is

completed independently.

STUDENT LESSON · 11 OF 18

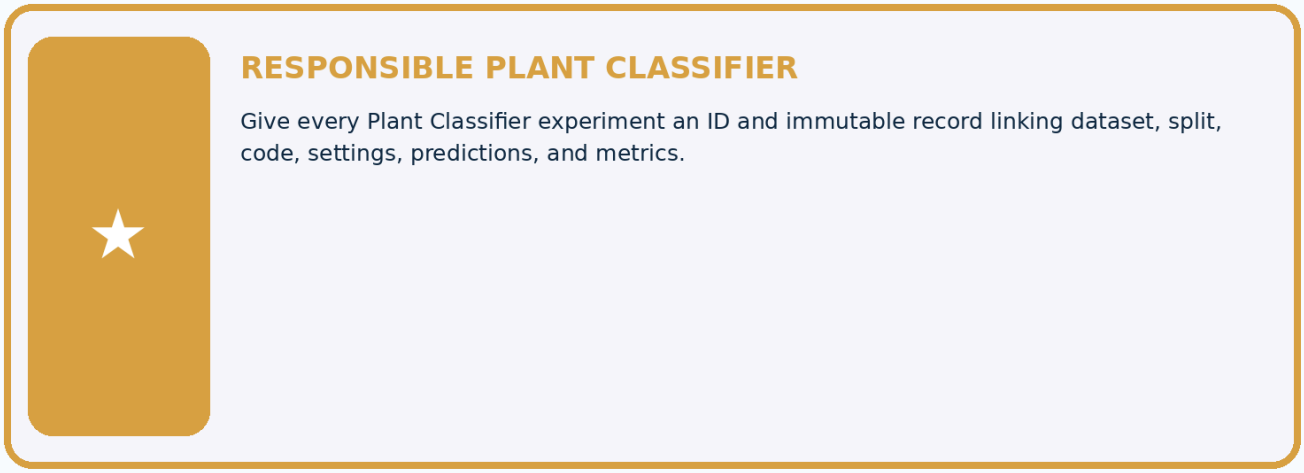
Connect your project

> Apply this week's idea to Responsible Plant Classifier. Keep an annotated decision and the evidence supporting it.

CUMULATIVE PROJECT

Responsible Plant Classifier

Every week adds one reviewable artifact to the same **evidence**-based capstone.



THIS WEEK'S CHECKPOINT Give every Plant Classifier experiment an ID and immutable record linking dataset, split, code, settings, **prediction**s, and metrics.

Which requirement or risk does this artifact address?

Which dataset, split, code, and metric **version** does it use?

What normal and shifted cases were tested?

What remains uncertain or weak?

What decision still belongs to a teacher or qualified human?

RESPONSIBLE RELEASE Never use this classroom prototype for medical, pesticide, agricultural, or other consequential decisions.

My **evidence**-linked project decision:

Evidence for your project

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

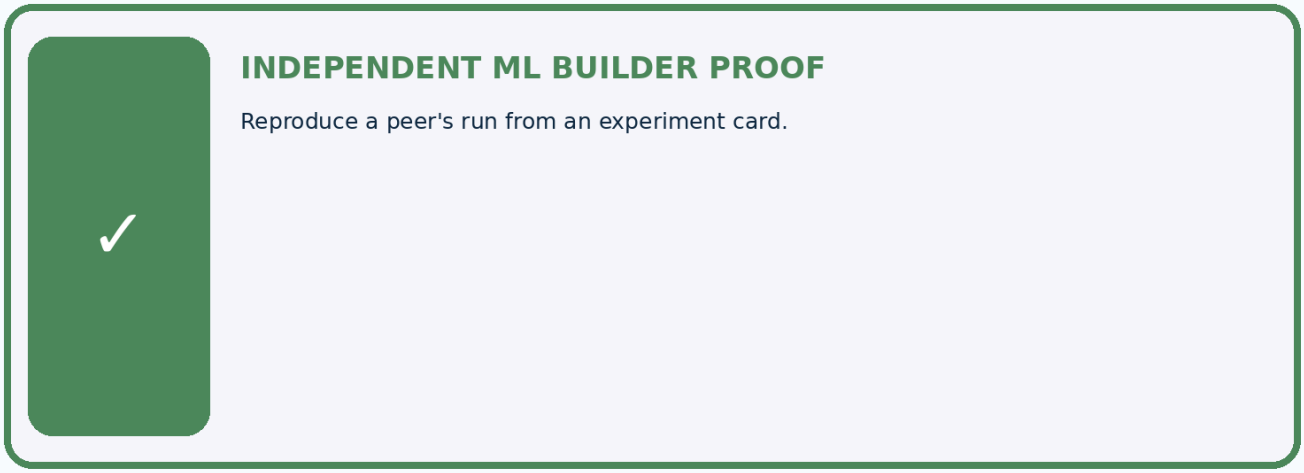
Show what you know

> Use a genuinely new case for independent evidence. Previously discussed examples remain practice; use the separately named independent case before feedback.

INDEPENDENT PROOF

Explain • Calculate • Transfer • Defend

Complete this with a fresh case after guided practice and compare it with the mastery criteria.



What success looks like

REQUIRED PROOF Reproduce a peer's run from an experiment card.

- ☐ Define the task, feature/target roles, and intended-use boundary.
- ☐ Show the calculation, code trace, or **experiment record**.
- ☐ Test a normal case and one revealing boundary, missing, or shifted case.
- ☐ Diagnose one failure without changing several factors at once.
- ☐ Name a limitation, fairness concern, and the human-review action.

Independent **evidence** and defense:

My level: ☐ Not yet ☐ With one hint ☐ Independently ☐ I can teach and defend it

> **Connect to the next lesson**
>
> NEXT BRIDGE Carry the experiment **evidence** into: Over**fitting**.

STUDENT LESSON · 13 OF 18

Study a complete case

> Revisit the complete Reproducible Experiments case for supported practice. Repeated examples do not become fresh assessments.

Week 26 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

A report says accuracy85% without data split, code **version**, **seed** or pre**process**ing.

The question

Specify **reproducibility** records.

Worked explanation

Record data identity, group/split IDs, code/dependency **version**s, pre**process**ing, features, settings, **seed**s where relevant, metric definition and raw **prediction**s. **Seed** alone is insufficient.

Explain it in your own words

> Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

Trace the supplied case visually

Make an experiment reproducible

Evidence or stage	Value or result
Data	Identity and split/group IDs
Code	Version and dependencies
Procedure	Features, preprocessing and settings
Evidence	Seeds, metric definitions and raw predictions

A report says accuracy85% without data split, code **version**, **seed** or pre**process**ing.

Specify **reproducibility** records.

Record data identity, group/split IDs, code/dependency **version**s, pre**process**ing, features, settings, **seed**s where relevant, metric definition and raw **prediction**s. **Seed** alone is insufficient.

Try a new case independently

> Attempt the reserved new case before feedback. Record support; the earlier examples below remain additional practice.

Independent case: Apply independently

Declared **seed**s 1, 2, 3 produce 9/12, 8/12, 10/12 correct. Compute the mean accuracy and range.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Week 26 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

> Two runs use different random splits and different feature sets. Their scores differ. What must be recorded, and why is the cause unclear?

My answer, with a trace, calculation or supporting **evidence**

Write or draw your reasoning and evidence.

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and expected result

> **Write or draw your reasoning and evidence.**

Connect to your project

Give every Plant Classifier experiment an ID and immutable record linking dataset, split, code, settings, **prediction**s, and metrics.

The **evidence** I will keep

> **Write or draw your reasoning and evidence.**

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Transfer practice from the original programme

Two teams use the same **random seed** but different label files. Their results differ. List the records needed to reproduce the experiment.

> **Explain the mechanism, evidence and one limit.**

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

STUDENT LESSON · 15 OF 18

Your lesson route

> Use this map to navigate Reproducible Experiments, connect the goals to tasks and keep one evidence record for each attempt.

Start with this question: Record variation across declared **seed**s instead of selecting the prettiest run.

By the end, I can	How I will show it
Record enough detail to reproduce a run	Explain the deciding step, show a trace or test, and state one limit.
Distinguish reproducibility from validity	Explain the deciding step, show a trace or test, and state one limit.
Report all planned seeds instead of selecting the best	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

STUDENT LESSON · 16 OF 18

Finish and return

> Explain Reproducible Experiments without notes, identify a remaining question and plan a later check with a changed case.

Close your notes. Explain this goal in your own words: Record enough detail to reproduce a run

> **My deciding step, evidence and one remaining question**

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

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Supported practice, repair and portfolio

> Attempt supported questions before using hints or feedback. Review a missing building block, then keep the completed work.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Record variation across declared **seed**s instead of selecting the prettiest run.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Record **data version**, split, code, settings and **seed**s, not just the winning**80%**.

1. Preserve context

2. Keep run identities

3. Summarise variation

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Preserve context Record **data version**, split, code, settings and **seed**s, not just the winning**80%**. Read the picture in order.

Record **data version**, split, code, settings and **seed**s, not just the winning**80%**. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: The report deletes the **60%** run. Which conclusion is supported by the supplied facts?

1. The remaining mean rises to75%, but the reported comparison becomes selectively incomplete.

2. This one example guarantees correct results for every future **input**.

3. We can decide the result without examining the changed **input** or the procedure.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: The remaining mean rises to75%, but the reported comparison becomes selectively incomplete.

Removing an unfavourable declared run changes the **evidence** rather than improving the method.

Repair a missing building block

Review Metrics Match Consequences (ai-4/25). Goal: Apply a declared score threshold including equality. Short practice: Which labels determine the false-alarm count at threshold 0.4?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Hold Out Test Data (ai-4/23). Goal: Freeze the full evaluation procedure before testing. Short practice: What is needed for an honest final claim about the revised **version**?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Mean, Median, and Baselines (ai-4/20). Goal: Fit mean and median constant baselines on **training** targets. Short practice: Does either original baseline win under evaluation MAE?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

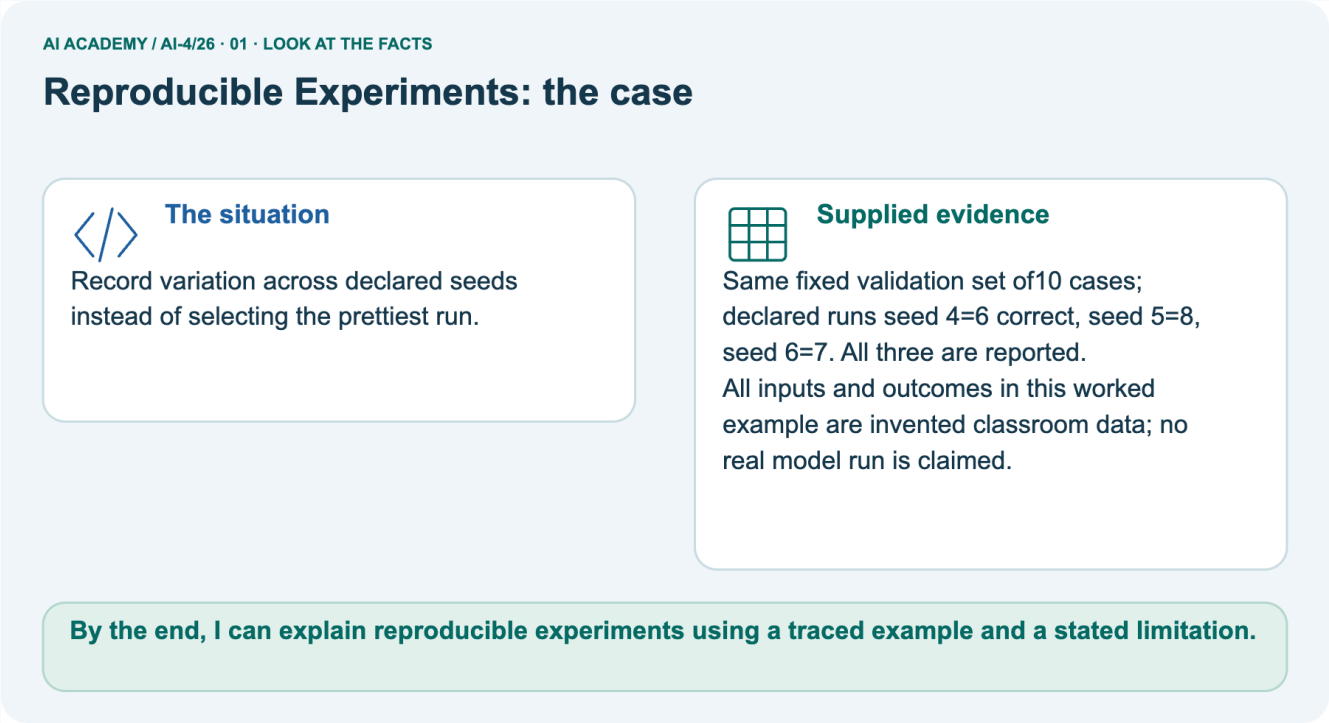
Drafts stay in this browser when storage is available; clear drafts on a shared device.

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Visual concept journey

> Explain each Reproducible Experiments graphic using its labels, deciding evidence and relevant limitation.

01. Reproducible Experiments: the case



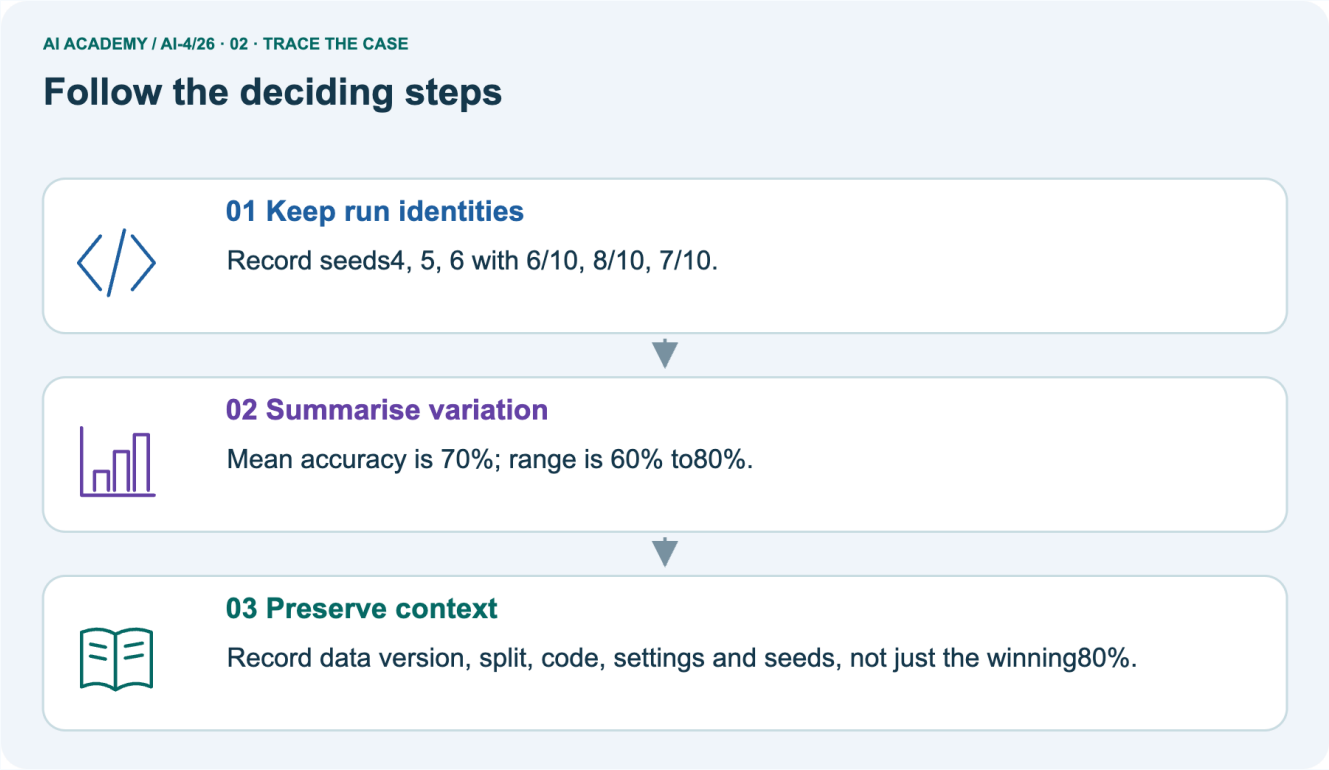
Follow the labelled picture with the matching explanation. Open the vector diagram

Record variation across declared **seed**s instead of selecting the prettiest run. Supplied facts: Same fixed validation set of10 cases; declared runs **seed** 4=6 correct, **seed** 5=8, **seed** 6=7. All three are reported.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

> Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Keep run identities: Record **seed**s4, 5, 6 with 6/10, 8/10, 7/10. → Summarise variation: Mean accuracy is **70%**; range is **60%** to**80%**. → Preserve context: Record **data version**, split, code, settings and **seed**s, not just the winning**80%**.

> Point to each numbered stage and explain which recorded **input** it uses.

03. All declared seeds

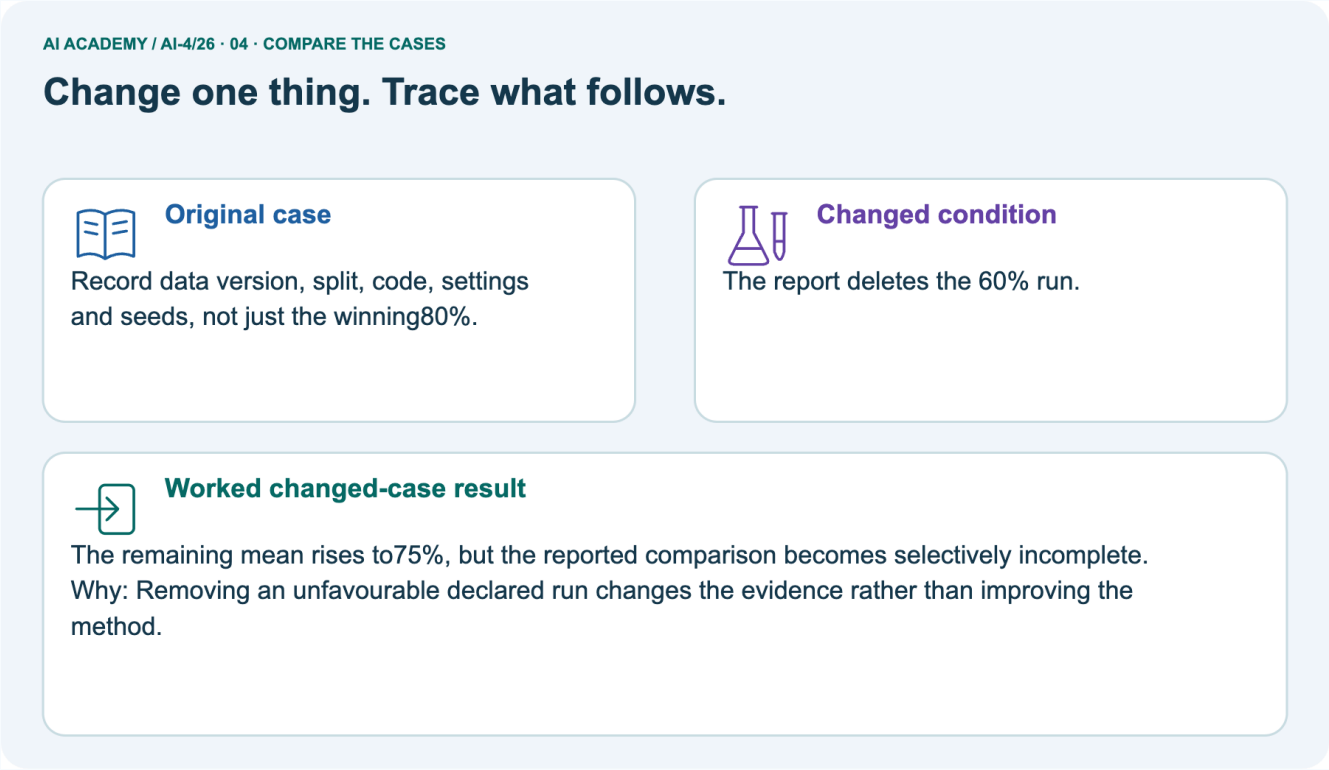


Follow the labelled picture with the matching explanation. Open the vector diagram

All declared **seed**s. Units: accuracy percent. **seed** 4: 60. **seed** 5: 80. **seed** 6: 70.

Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



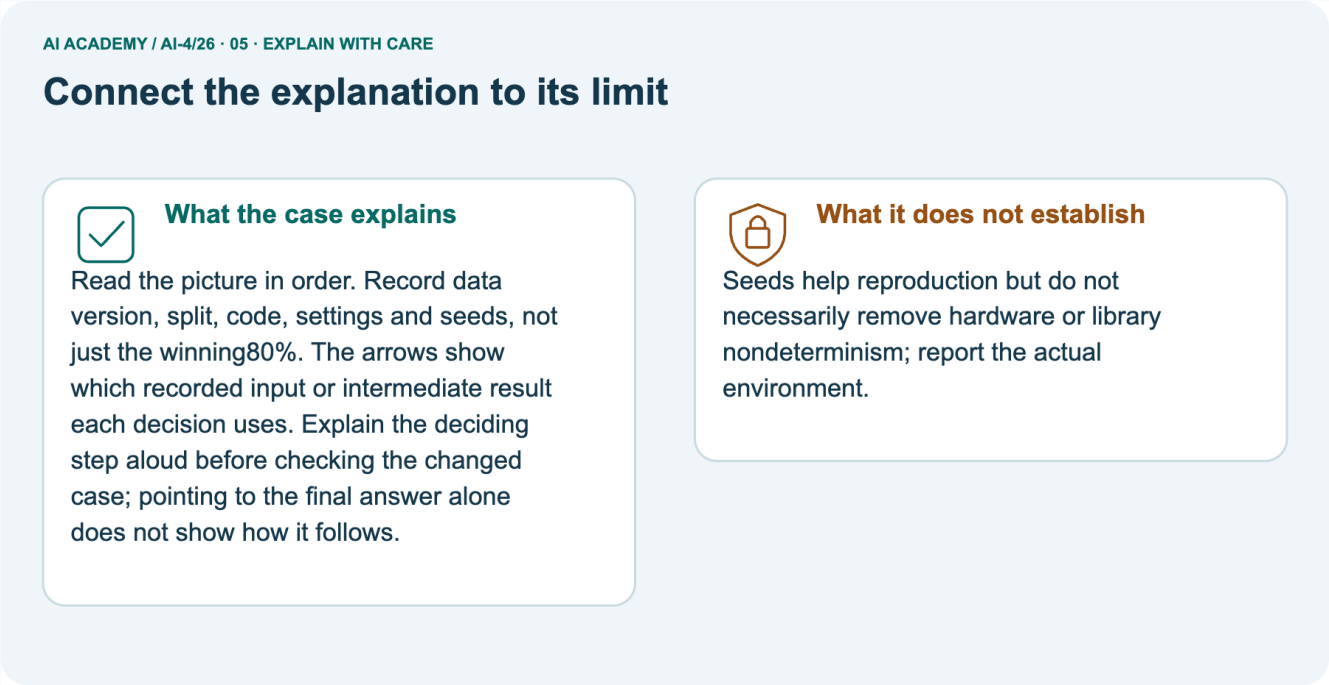
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Record **data version**, split, code, settings and **seed**s, not just the winning**80%**. Changed condition: The report deletes the **60%** run.

Expected result: The remaining mean rises to75%, but the reported comparison becomes selectively incomplete. Reason: Removing an unfavourable declared run changes the **evidence** rather than improving the method.

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

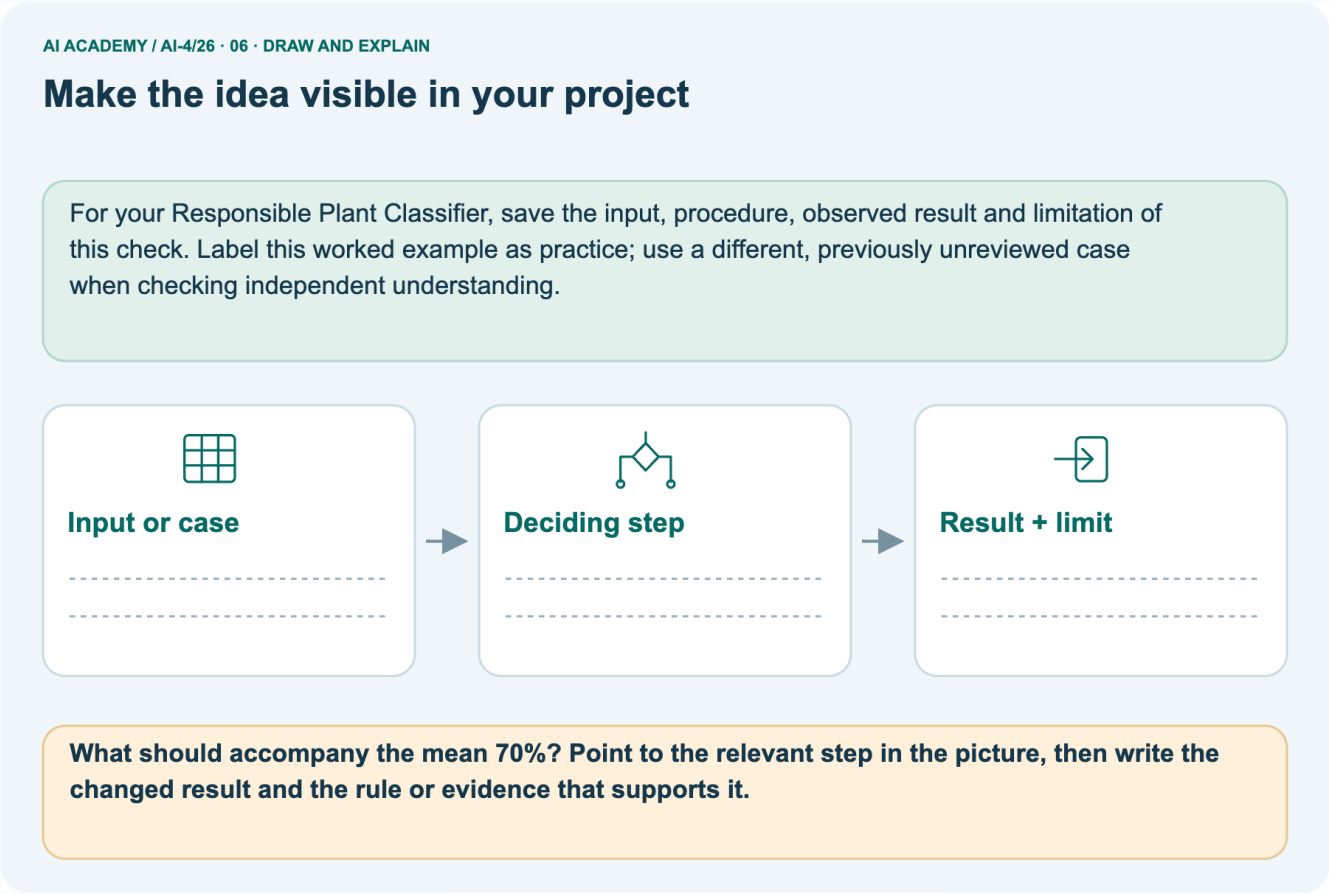
Read the picture in order. Record **data version**, split, code, settings and **seed**s, not just the winning**80%**. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Limit: **Seed**s help reproduction but do not necessarily remove hardware or library nondeterminism; report the actual environment.

> Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: What should accompany the mean **70%**? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> Draw your project trace and justify the deciding step using **evidence**.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.